Track 2 — Mechanism and Drug Repurposing for a *BUB1B* Mosaic Variegated Aneuploidy Case

Team PHA (Pari, Aradhya, Harshita) · Rare Disease, Real Kid: MVA Hackathon 2026 Code and data provenance: `track2/pipeline/` in github.com/pari25mib10005-byte/mva-hackathon-2026-track1

> **Everything in this report is a hypothesis for investigation, not treatment advice.** Any use in the child must be decided by his clinicians under appropriate ethical oversight.

At a glance

- **Mechanism:** compound-heterozygous *BUB1B* loss-of-function (hypomorph): too little BUBR1, weak spindle checkpoint, mosaic aneuploidy and cancer risk. Supported by LOFTEE (truncating allele) and AlphaMissense 0.92 (missense allele).
- **Search:** three independent routes (gene-, phenotype-, transcriptome-first) agree the checkpoint cannot be safely drugged directly.
- **Lead:** NAD^+ repletion via SIRT2-mediated BUBR1 stabilisation: **niacin** (approved medication) or **nicotinamide riboside** (paediatric safety data).
- **This child:** both drugs' conversion pathways intact; vincristine-neuropathy risk genotypes; Factor V Leiden; a testable second-diagnosis candidate (*PEX5*).
- **Status:** hypotheses for investigation, with a staged validation plan; not treatment advice.

In plain language

This child has a rare condition, mosaic variegated aneuploidy (MVA), in which some of his cells end up with too many or too few chromosomes when they divide. We found that it is caused by two faulty copies of a gene called *BUB1B*. That gene makes BUBR1, a protein that acts like a safety check, telling a cell to wait until its chromosomes are sorted correctly before dividing. His cells still make some BUBR1, but too little, because it is broken down too quickly.

Working backwards from his symptoms alone, our analysis independently pointed to the same gene and the same faulty cell-division safety check. We then looked for a medicine that already exists and might help his cells keep more of the BUBR1 they make. Our best idea is to raise a natural molecule called NAD^+, which helps the cell protect BUBR1. Two forms of vitamin B3 can do this: **niacin**, an approved prescription medicine, and **nicotinamide riboside**, a supplement that has been given to children with another rare condition. In mice, this approach increased BUBR1. It has **not** been tested for MVA. It might reduce new chromosome errors over time, but it cannot undo features that formed before birth. We set out how it should be tested in the lab before anyone considers it for him.

Summary

Our Track 1 analysis identified compound heterozygous *BUB1B* variants (one truncating, one missense; exact variant identifiers are withheld from public materials). We argue that this genotype produces a **BUBR1 hypomorph**: the child's cells make some BUBR1, but too little, because one allele is null and the other encodes an unstable protein. The therapeutic goal is therefore to **stabilise the BUBR1 that remains**, while avoiding anything that weakens the mitotic checkpoint further or adds cancer risk.

A systematic, gene-agnostic screen of DGIdb and Open Targets across 29 related genes found **no usable drug that targets BUB1B** and **no approved drug known to push the BUBR1-stability pathway in the right direction**; all 552 drug–gene records were excluded as harmful, wrong-direction, unsafe or investigational-with-unknown-direction, except one approved compound of unknown direction. An independent phenotype-driven analysis, starting from the patient's clinical features rather than his DNA, ranked *BUB1B* 16th of 2,479 genes and converged on the same checkpoint pathway, with no directly druggable target in the helpful direction. A third, pathway-agnostic route (LINCS L1000 signature reversal) tested the NAD^+ hypothesis with a pre-specified statistic: support was weak and not significant ($p = 0.085$), and the analysis showed that naive reversal of a chromosome-instability signature favours drugs that silence the cell's protective p53/interferon alarm (93 of the top 100), which we exclude. The lead strategy is therefore **NAD^+ repletion**, with two options: **niacin**, an approved prescription medicine that raised blood NAD^+ up to 8-fold in humans, and **nicotinamide riboside (NR)**, a supplement with the stronger paediatric safety record. Local checks of the child's genome show that both drugs' conversion pathways are intact, flag two genetic risk factors for vincristine neuropathy and Factor V Leiden, and raise one possible second diagnosis (*PEX5*) that a standard blood test can confirm or exclude. Variant-effect predictors support the mechanism (AlphaMissense 0.92, likely pathogenic, for the missense allele; high-confidence loss-of-function for the truncating allele). Both act on the NAD^+ → SIRT2 → BUBR1 axis that raised BubR1 levels in mice. We expect at most a partial benefit and propose a staged validation plan before any clinical use.

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1. Mechanism

Mechanism classification: loss-of-function (hypomorphic). Pathway disrupted: **spindle assembly checkpoint and**

kinetochore–microtubule attachment. Downstream consequence: **chromosome missegregation → mosaic random aneuploidy → growth/developmental features and cancer predisposition**.

1.1 Starting point

Track 1 (scored 100/100; F-max 1.000) identified two *BUB1B* variants in trans:

Allele	Variant	Type	Predicted effect
A	Truncating (premature stop codon)	Nonsense	Null (nonsense-mediated decay)
B	Single amino-acid substitution	Missense, kinase domain	Full-length but unstable protein

Biallelic *BUB1B* mutations, typically one truncating plus one missense allele, are an established cause of MVA [2]. The mechanism below is therefore anchored in a confirmed genotype, not a guessed one.

1.2 What BUBR1 does

BUBR1 is a core component of the spindle assembly checkpoint (SAC). It has two jobs:

1. **Checkpoint signalling.** Its N-terminal KEN boxes and TPR region bind CDC20 and, within the mitotic checkpoint complex, keep APC/C–CDC20 inactive until every kinetochore is attached [3].
2. **Attachment stability.** A central region recruits PP2A-B56 to kinetochores, stabilising correct kinetochore–microtubule attachments [3].

The C-terminal (pseudo)kinase domain is largely dispensable for catalysis in vertebrates, but its structural integrity prevents degradation of the whole protein [4].

1.3 Predicted effect of each allele

Allele A (truncating). Truncating *BUB1B* alleles in MVA patients give no detectable transcript, consistent with nonsense-mediated decay [1]. Any truncated protein that escaped NMD would lack the stabilising C-terminal domain [4]. We model this allele as null.

Allele B (kinase-domain missense). This introduces a charge-altering substitution inside the kinase domain. In patient cells, MVA missense mutations in or near this domain consistently caused rapid BUBR1 turnover rather than loss of a specific function [1]. We predict a full-length but unstable protein.

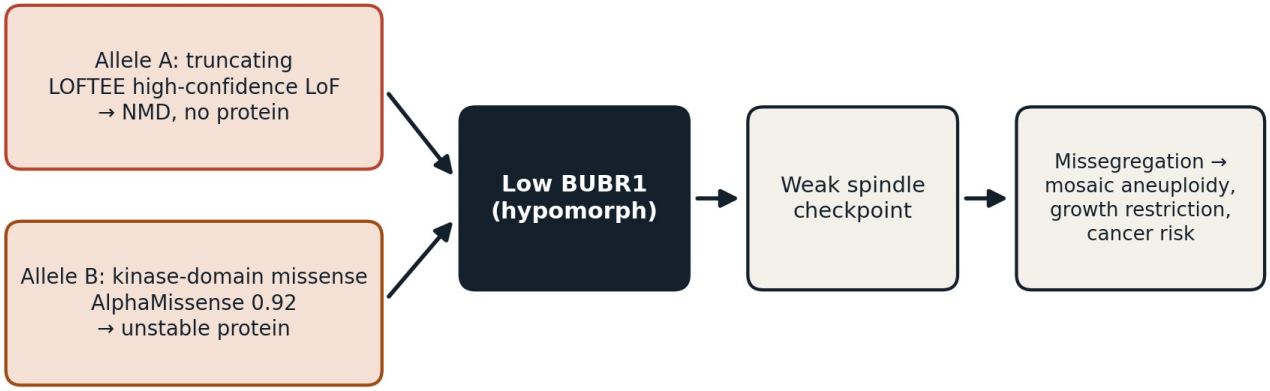
Variant-effect predictors (Ensembl VEP, October 2026) support both predictions:

Predictor	Allele A (truncating)	Allele B (kinase-domain missense)
LOFTEE	**High-confidence loss-of-function**; 50-bp rule passes, so nonsense-mediated decay is predicted [27]	n/a
AlphaMissense	n/a	**0.923, likely pathogenic** (threshold 0.564) [23]
SIFT / PolyPhen-2	n/a	Deleterious (0.01) / probably damaging (0.997)
CADD (phred)	36	24.5 [24]
SpliceAI (max delta)	0.03	0.00 [25]

The missense allele is predicted to damage the protein without affecting splicing, consistent with a full-length but unstable BUBR1; the truncating allele is predicted to be removed by NMD. Exact variant identifiers are withheld.

1.4 Combined model: a BUBR1 hypomorph

Figure 1. Mechanism: a loss-of-function BUBR1 hypomorph



Therapeutic goal: stabilise the BUBR1 that remains, without weakening the checkpoint or adding genotoxic risk

Genotype to phenotype: the BUBR1 hypomorph model.

```
Allele A (null, NMD) + Allele B (unstable protein)
        ↓
Low total BUBR1 protein (reduced, not absent)
        ↓
Weakened SAC + less stable kinetochore–microtubule attachment
        ↓
Chromosome missegregation in dividing cells
        ↓
Mosaic random aneuploidy → growth/developmental features + cancer predisposition
```

This places the case in the "low protein abundance" class of MVA mutations [1]. Reintroducing BUBR1 into MVA patient cells restored checkpoint activity [1], so the cellular defect is in principle reversible by raising BUBR1 levels. The BubR1 hypomorphic mouse (BubR1^H/H) is the closest animal model: low BubR1, aneuploidy, progeroid features and shortened lifespan [7].

1.5 From mechanism to drug strategy

No drug targets BUBR1 directly, so we looked for ways to **slow its degradation**. The key published observation [5]:

- With age, BubR1 is lost because NAD^+ falls and SIRT2 can no longer keep **lysine-668** deacetylated; the acetyltransferase **CBP** acetylates this residue, priming BubR1 for degradation.
- Raising NAD^+ with **nicotinamide mononucleotide (NMN)**, or overexpressing SIRT2, **increased BubR1 abundance in vivo**. SIRT2 overexpression also extended median lifespan of BubR1^H/H mice [5, 6].
- SIRT2 did not restore normal BubR1 levels or remove all progeroid features [5, 6], so any rescue should be expected to be partial.

This defines the **NAD^+ → SIRT2 → BUBR1 stabilisation axis** as the primary search space, and gives four design rules used throughout:

1. Raise NAD^+ / SIRT2 activity, or otherwise slow BUBR1 turnover.
2. Never weaken the checkpoint or increase missegregation.
3. Avoid genotoxic agents (MVA predisposes to cancer [2]).
4. Prefer agents with paediatric safety data.

1.6 Independent lines of support for the lever

The NAD^+ → SIRT2 → BUBR1 lever does not rest on a single study. Four independent kinds of evidence point the same way; what remains untested is only the final link in human cells, which is the first experiment of the validation plan (Section 10).

Evidence line	Model	Finding	Ref
NAD^+/SIRT2 stabilise BubR1	Mouse (in vivo)	NAD^+ precursor raised BubR1; SIRT2 overexpression extended lifespan of BubR1-hypomorphic mice	[5]
SIRT2 protects chromosome fidelity	Sirt2-knockout mouse, human tumours (different laboratory)	SIRT2 loss caused aneuploidy and tumours; SIRT2 acts as a tumour suppressor via mitotic regulation, so boosting it should oppose, not fuel, chromosomal instability	[33]
NAD^+ precursors raise NAD^+ in humans	Human trials	Niacin raised blood NAD^+ up to 8-fold; NR raised NAD^+ in children and adults	[26, 8, 9]
NR in human genome-instability syndromes	Randomised placebo-controlled trial (Werner syndrome) and open-label studies (ataxia-telangiectasia, children from age 3)	NR raised NAD^+ and improved clinical markers without serious adverse events; liver-test changes were noted, supporting liver monitoring	[34, 8, 9]
This child's genome	Local VCF analysis	NAD^+-conversion genes and *SIRT2* intact (Section 7.1)	—

2. Systematic candidate search (reproducible pipeline)

To show the candidate list is not hand-picked, we built a **gene-agnostic pipeline** (`track2/pipeline/repurpose.py`, Python standard library only). It works for any gene through a config file; we tested it unchanged on the other MVA genes, *TRIP13* and *CEP57*.

2.1 Method

1. **Gene set (29 genes):** *BUB1B*; curated modifiers with the direction the mechanism needs (SIRT2 ↑, NAMPT ↑, CREBBP ↓, EP300 ↓); and high-confidence STRING neighbours (score ≥ 0.9) from Open Targets.
2. **Evidence:** drug–target records from **DGIdb** and **Open Targets** (552 raw records; live query run on 19 October 2026).
3. **Hard exclusion** by target or mechanism class (shared config `cin_exclusions.json`): antimitotics/tubulin, Aurora, PLK1, MPS1/TTK, CDK1, KIF11/CENPE, PP2A, proteasome, HSP90, topoisomerases, DNA-synthesis inhibitors, PARP.
4. **Direction filter:** a drug must move each modifier in the needed direction; for a loss-of-function gene, inhibitors of the gene or its network partners are rejected.
5. **Whole-drug safety pass:** every surviving drug is re-checked against its *full* Open Targets mechanism profile, so a hazardous drug cannot enter through a side annotation.
6. **Chemotherapy-by-name pass:** every drug in WHO ATC group L01 (antineoplastic agents, via ChEMBL) is excluded by name, so a chemotherapy cannot enter when a database record lacks its mechanism or uses a variant name (e.g. a prodrug).
7. **Curated wrong-direction mechanisms:** research compounds that the databases do not annotate (e.g. sirtuin and NAMPT inhibitors) are excluded from a documented list in the config; investigational compounds with unknown direction are not proposed for a child.
8. **Not-proposable modifiers:** CREBBP/EP300 inhibition is itself unsafe (CREBBP loss causes Rubinstein–Taybi syndrome; both are tumour suppressors), so these genes are used only to exclude activators.
9. **Curated literature layer:** agents the databases cannot represent (metabolic precursors) are added and labelled as such.
10. **Outputs:** ranked candidates, a full exclusion log with reasons, and a provenance file (run time, sources, config, gene set).

2.2 Results

Measure	Result
Drug records on *BUB1B* itself	**1** (a gemcitabine prodrug from a cell-sensitivity screen; excluded as chemotherapy)
Raw drug–gene records across the gene set	552
Excluded as harmful or wrong direction	**474**
Set aside: CBP/p300 inhibitors (right direction, unsafe)	28
Set aside: investigational compounds with unknown direction	49
Database hits with a confirmed, correct direction on the BUBR1 axis	**0**
Approved compounds with unknown direction (manual review)	1

Largest exclusion groups: PLK1 inhibitors (199), CDK1 (109), Aurora B (103), NAMPT inhibitors (11, would lower NAD^+), PP2A inhibitors (7), sirtuin inhibitors (4). Successive safety passes caught agents that would otherwise have slipped through: **colchicine** (a tubulin poison listed in DGIdb against CREBBP), Aurora inhibitors listed against NEK2, and **four cancer drugs** (a gemcitabine prodrug, gefitinib, palbociclib, pazopanib) whose records lacked a usable mechanism. Counts come from live databases and drift between runs; earlier runs gave 513 records and the same conclusion.

Interpretation. Drug–target databases contain no approved drug that stabilises BUBR1, and the obvious "related" drugs are almost all harmful in this setting. The NAD^+ precursors are absent because they are metabolic substrates, not target binders. The lead candidate therefore comes from the literature layer, after systematic search and filtering.

3. Phenotype-driven analysis (independent evidence line)

Section 2 starts from the gene. This analysis starts from the **patient's clinical features** (HPO terms, used locally and not reproduced here) and asks which genes and shared pathways explain them, without using any sequencing data (`track2/phenotype/phenotype.py`).

Method. Every gene in the HPO gene–phenotype table (5,276 genes) was scored by the summed information content of the patient terms it explains, so specific features outweigh common ones. A STRING network (combined score ≥ 0.7) was built over the top 200 phenotype genes plus the disease genes; the causal gene's direct partners define the shared-biology module, tested for pathway enrichment (Reactome, KEGG, GO). Drugs acting on the module were retrieved and passed through the same exclusion, direction and whole-drug filters as Section 2. One family-history term was excluded from the patient's phenotype.

Result	Value
Rank of *BUB1B* from phenotype alone	**16 of 2,479** matching genes (top 0.6%); *CEP57* 17th, *TRIP13* 104th
Direct network partners of *BUB1B* among phenotype genes	BUB1, BUB3, TRIP13, PPP2R5C (PP2A-B56), TP53
Top enrichment of the module	KEGG cell cycle (FDR 5.9×10^{-10}); GO mitotic spindle assembly checkpoint signalling (FDR 2.4×10^{-7}); Reactome amplification of signal from unattached kinetochores (FDR 7.0×10^{-6})
Patient features explained by MVA genes	3 of 7; **4 of 7 are not annotated to any MVA gene**
Drugs acting on the module with known action, correct direction and passing safety filters	4, all via TP53, none proposable (below)
Association-only links set aside (e.g. TP53 used as a trial biomarker)	466

Interpretation.

- The phenotype alone recovers the Track 1 gene near the top of 2,479 candidates, an independent validation of the genetic diagnosis.
- The biology shared between the patient's features and *BUB1B* is the **spindle assembly checkpoint**, including PP2A-B56, which matches the mechanism in Section 1.
- The four remaining module drugs do not fit this patient: eprenetapopt restores *mutant* p53 (no TP53 defect is implicated here), two are cancer vaccines, and one is a database artefact. The checkpoint core therefore cannot be drugged directly in the helpful direction, which independently supports the indirect, stabilisation strategy.
- Four of the patient's features are not explained by MVA gene annotations. Section 3.1 tests whether they are genetic.

3.1 Genetic or not: which features does the MVA network explain?

Method. *BUB1B*'s 100 strongest STRING partners (score ≥ 0.7), selected without reference to the patient's features (30 carry HPO annotations), were tested for each feature with a hypergeometric test (universe 5,276 HPO genes). Features are reported here by category only.

Feature category (n)	Annotated to an MVA gene	Enrichment among *BUB1B* partners	Classification
Tumour (1)	Yes	4 vs 0.1 expected, **$p = 5\times10^{-6}$**	Genetic (MVA)
Growth before and after birth (2)	Yes	17 vs 8.4, **$p = 8\times10^{-4}$**; 3 vs 1.7, p = 0.25	Genetic (MVA)
Perinatal (1)	No	0 vs 1.2, p = 1.0	Likely secondary: growth-restricted fetuses are often delivered early
Nutritional growth after birth (1)	No	4 vs 6.7, p = 0.93	Likely secondary: growth disorder plus illness
Muscle (1)	No	0 vs 3.4, p = 1.0	Likely acquired: nutrition, the cancer itself, or treatment (vincristine, standard for this tumour type, causes neuropathy and muscle wasting); the child carries genetic risk factors for vincristine neuropathy (Section 7.1)
Renal (1)	No	0 vs 0.6, p = 1.0	Acquired (e.g. loop diuretics in premature infants, some chemotherapies) **or a second diagnosis**

The MVA network explains the tumour and growth features strongly and the other four not at all, so those four are secondary, acquired, or due to a second cause. "Secondary" and "acquired" are hypotheses from known causes; treatment history was not available to us.

Possible second diagnosis. The renal, perinatal and nutritional features together form the classic pattern of antenatal Bartter syndrome (a salt-wasting tubulopathy), and the phenotype scoring independently ranked two of its genes, *SLC12A1* and *KCNJ1*, first and second of all genes. Dual molecular diagnoses occur in a minority of solved cases [22]. `track2/tools/check_vcf_genes.py` checks the patient VCF **locally** for rare variants in *SLC12A1, KCNJ1, CLCNKB, BSND* and *MAGED2* and prints only gene-level verdicts. **Result (local run, 19 October 2026): no rare protein- or splice-altering variants in any of the five genes** (three rare non-coding variants, low prior). A second genetic cause of this tubulopathy type is therefore not supported by the small-variant calls,

which favours an acquired explanation for the renal and perinatal features. Caveat: the VCF holds small variants only, so whole-gene deletions (a common cause of *CLCNKB*-related Bartter syndrome) and deep-intronic effects are not assessed; a copy-number review or a clinical tubular work-up remains reasonable if the history does not explain the findings. Clinical history (diuretics, chemotherapy, urinary calcium) should be reviewed first.

3.2 Genome-wide check: is there a second genetic cause?

Section 3.1 tested only the disease-gene network. To test "genetic or not" properly, `tools/check_feature_genes.py` took **every gene known to cause each of the four unexplained features** (HPO; 1,700+ genes in total, HLA genes excluded as unreliable), scanned their coding exons (MANE Select, ±10 bp) in the patient's VCF locally, and applied inheritance-aware rules: recessive genes need two rare damaging hits, dominant genes one ultra-rare hit. "Damaging" was judged on each gene's **canonical transcript** (loss-of-function, AlphaMissense likely-pathogenic missense, or a novel in-frame indel), with ClinVar-benign variants excluded. Two earlier false alarms (HLA hypervariability; frameshifts on minor transcripts) were removed by these rules.

Unexplained feature (category)	Genes scanned	Result
Renal	106	**No gene with a disease-consistent pattern**: a second genetic cause is not supported; acquired cause favoured
Nutritional growth after birth	1,176	*PEX5* (homozygous, see below); *SERPINA1* (two heterozygous frameshifts of near-identical population frequency, consistent with one complex variant on the same copy; low prior)
Muscle	598	*PEX5*; plus vincristine-neuropathy risk genotypes (Section 7.1)
Perinatal	203	*PEX5*

A possible second diagnosis: *PEX5*. The patient is homozygous for a variant at the splice donor of *PEX5*'s canonical transcript, absent from gnomAD and from ClinVar. If it disrupts splicing, biallelic *PEX5* loss causes a peroxisome biogenesis disorder (Zellweger spectrum), which could link three of the four unexplained features. **Call quality supports it:** all 10 reads at the site carry the variant and none the reference allele, the call passes filters (QUAL 417, genotype quality 37). Depth is lower than typical for a whole genome, which is common at splice-site indels; reduced depth could also reflect a deletion of the other copy, which would still leave both copies affected. We therefore report a **well-supported candidate requiring confirmation**: a homozygous novel variant needs orthogonal (Sanger) confirmation, and the classic severe phenotype includes features not listed for this patient. It is cheap to resolve: **plasma very-long-chain fatty acids** (with plasmalogens, phytanic and pristanic acid), Sanger confirmation of the variant, and parental carrier testing. *SERPINA1* can be settled with a serum alpha-1 antitrypsin level.

3.3 Family history: recurrent pregnancy loss

The family-history term has no overlap with MVA genes or *BUB1B* partners, but the child's genome still informs it:

- **Factor V Leiden is present in the child**, so one parent carries it. If the mother carries it, this inherited thrombophilia is a recognised contributor to recurrent pregnancy loss (prothrombin G20210A was absent in the child) [31]. Maternal thrombophilia has also been linked, inconsistently, to placental complications such as growth restriction and preterm birth, a possible contributor to the perinatal features.
- **BUB1B carrier status:** a compound-heterozygous child implies both parents are carriers, giving a 25% recurrence risk per pregnancy. An affected pregnancy would carry the same two alleles as the child, whose survival shows this genotype is compatible with life, so *BUB1B* sets the recurrence risk but is an unlikely cause of the losses themselves.
- **Common non-genetic causes remain the most likely overall:** chance chromosome errors in the embryo, antiphospholipid syndrome (the most treatable cause) and uterine factors; a standard recurrent-pregnancy-loss work-up for the mother addresses these.
- Both are testable: parental Factor V Leiden and *BUB1B* carrier testing, with genetic counselling. The finding also matters for the child himself: Factor V Leiden raises venous clot risk, relevant to central lines and immobility during cancer care.

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4. Transcriptomic signature reversal (third, pathway-agnostic route)

Routes 1 and 2 both use prior knowledge of genes and pathways. This route asks, with **no pathway assumption**, which compounds push human cells in the opposite direction to losing *BUB1B* (`track2/lincs/signature_reversal.py`).

Method. The LINCS L1000 CRISPR-knockout consensus signature of *BUB1B* (247 genes up, 244 down; no signatures exist for *CEP57* or *TRIP13*) was compared with consensus signatures of 5,425 compounds [18, 19]. Score = (reversing − mimicking overlaps) / signature size; direction tested with a binomial sign test, Benjamini–Hochberg FDR. **Pre-specified hypothesis, fixed before results:** NAD^+/sirtuin-axis compounds reverse the signature more than other compounds (one-sided Mann–Whitney U). Top reversers then passed the Section 2 safety rules, wrong-direction patterns (sirtuin, NAMPT, JAK and p53 inhibitors), and exclusion of every WHO ATC L01 antineoplastic [21], which catches chemotherapies whose database mechanism is missing (e.g. procarbazine, estramustine).

What the knockout signature contains. The "up" half includes the cell's **protective alarm** to chromosome errors (p53/apoptosis genes such as *CDKN1A*, *GDF15*, *BAX*; interferon genes such as *ISG15*, *MX1*, a known response to chromosome instability [20]); the "down" half includes **cell-division genes**, among them *BUB1B*, *BUB1*, *TRIP13*, *CDC20*, *CCNB1* and *PLK1*.

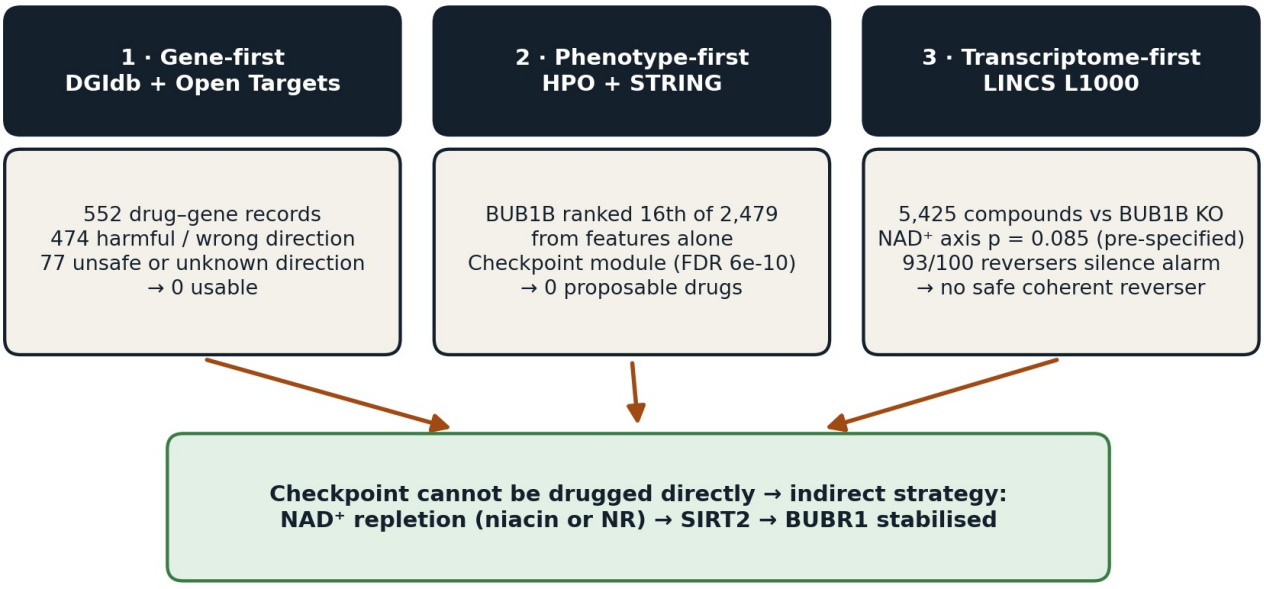
Result	Value
Significant reversers (FDR < 0.05)	1,134 of 5,425
NAD^+/sirtuin axis (pre-specified)	SRT-1720 rank 283 (FDR 7×10^{-4}); resveratrol 1,295 (FDR 0.10); NAD^+ (Nadide) 1,683 (FDR 0.15); nicotinamide 3,256 (FDR 0.97). **Group test p = 0.085 (not significant)**; NR and NMN are not in LINCS
Niacin (NAD^+ precursor; **not pre-specified**, exploratory)	Rank 98 (top 2%, FDR 2×10^{-6}); restores division genes (FDR 4×10^{-8}); switches off 3 alarm genes (*ATF3*, *CDKN1A*, *MX1*). Post-hoc group test including niacin: p = 0.024; NAD^+ precursors only: p = 0.13
Top 100 reversers that switch off ≥ 3 alarm genes	**93** (e.g. tofacitinib, a JAK inhibitor; EX-527, a sirtuin inhibitor; APO-866, a NAMPT inhibitor)
Top 100 remaining after all safety rules	6 (2 approved), with no coherent mechanism
Exploratory: restore division genes without silencing the alarm	100 at FDR < 0.05; 93 pass safety (27 approved), mechanistically heterogeneous (antipsychotics, local anaesthetics, hormones, NSAIDs)

Interpretation.

- **The NAD^+ hypothesis gains no significant transcriptomic support.** Only the synthetic SIRT1 activator SRT-1720 reverses significantly; nicotinamide does not, consistent with its sirtuin-inhibiting effect at high concentration (Section 7). Because L1000 measures transcripts, it cannot see NR's proposed mechanism (BUBR1 acetylation and protein turnover), so this is an absence of support, not evidence against. A human niacin trial found the same pattern: NAD^+ repletion corrected the metabolome without correcting transcriptional stress responses [26]. Niacin, omitted from our pre-specified list, is a top-2% reverser; we report this as exploratory, and its partial alarm suppression (*CDKN1A*) becomes a readout in the cell experiments.
- **Naive reversal is the wrong objective for chromosome-instability disorders.** Most strong reversers act by silencing the p53/interferon alarm, which would let aneuploid cells survive in a cancer-prone child. Reversal must be split into its two halves and alarm-suppressors excluded; we report this as a methodological finding.
- **No new candidate is promoted from the screen itself;** niacin enters the shortlist on mechanism (Section 7), with its LINCS result as exploratory support. The safe, division-restoring compounds share no pathway, These compounds are treated as exploratory comparators for the cell screen (Section 10), not as leads.

5. Convergence of evidence

Figure 2. Three independent routes converge



The three routes and their shared conclusion.

Question	Route 1: gene-first (Section 2)	Route 2: phenotype-first (Section 3)	Route 3: transcriptome-first (Section 4)	Verdict
Is *BUB1B* the right gene?	Track 1 genotype	Ranked 16th of 2,479 from features alone	Knockout lowers checkpoint genes (*BUB1*, *TRIP13*, *CDC20*, *PLK1*)	**Consistent**
Which biology is disrupted?	Spindle checkpoint; PP2A-B56 attachment	Module enriched for checkpoint signalling	Loss of division genes plus p53/interferon alarm	**Consistent**
Can it be drugged directly in the right direction?	0 direction-confirmed approved drugs	0 proposable module drugs	No safe reverser with a coherent mechanism	**Consistent (no)**
Is the NAD⁺/sirtuin strategy supported?	Mechanism-based (literature)	Indirect strategy required	Pre-specified p = 0.085; niacin top 2% (exploratory, post-hoc p = 0.024)	**Mechanism-based; suggestive, not confirmed, in transcripts**
Which features can a drug address?	n/a	Tumour and growth features (genetic); four others secondary, acquired or a second cause	n/a	**Scope defined**

Three independent starting points agree on the gene, the pathway and the conclusion that the checkpoint cannot be drugged directly. The choice of an indirect BUBR1-stabilising strategy rests on that agreement; the choice of NR within it rests on mechanism and paediatric safety, and remains a hypothesis for the validation plan (Section 10).

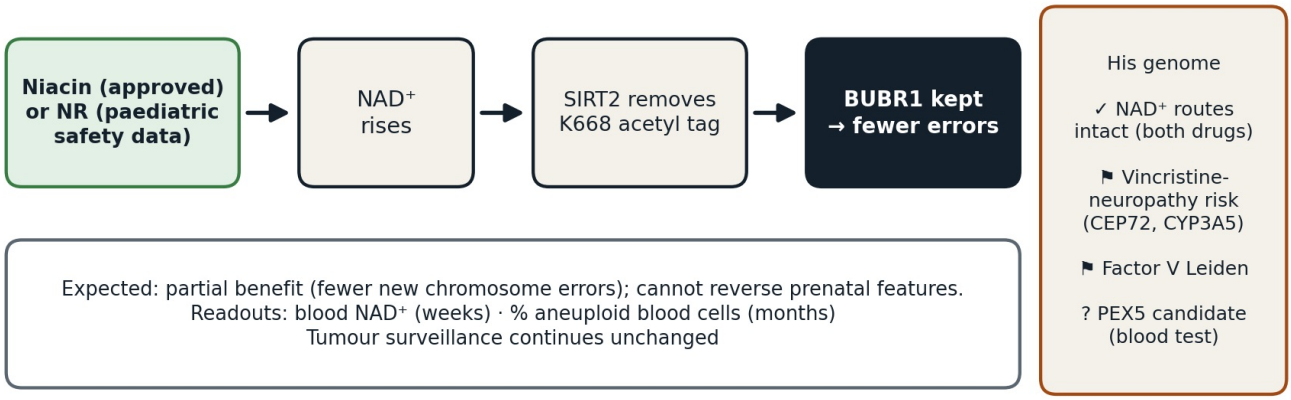
6. Exclusion filter

Agents predicted to worsen missegregation or cancer risk are excluded regardless of other merits.

Excluded class	Examples	Reason
Antimitotics	Taxanes, vinca alkaloids, colchicine	Disrupt spindles on top of a weak checkpoint
Mitotic kinase inhibitors	Aurora, MPS1/TTK, PLK1, CDK1 inhibitors	Weaken the same checkpoint and error-correction machinery that is already deficient
KIF11/CENPE inhibitors	Ispinesib, filanesib	Cause mitotic arrest and errors
PP2A inhibitors	Cantharidin	BUBR1 recruits PP2A-B56 to stabilise attachments
Proteasome inhibitors	Bortezomib, carfilzomib	Would slow BUBR1 degradation, but are cytotoxic with broad mitotic effects; unacceptable in a child
HSP90 inhibitors	Investigational agents	Destabilise many client proteins; not selective
Sirtuin / NAMPT inhibitors; CBP/p300 activators	Daporinad, AGK2	Predicted to speed BUBR1 acetylation and loss
CBP/p300 inhibitors	Inobrodib, GNE-049	Directionally consistent but unsafe (tumour suppressors; Rubinstein–Taybi)
Genotoxic agents	Alkylators, topoisomerase poisons, PARP inhibitors	MVA predisposes to cancer

7. Candidates

Figure 3. How the lead works, and what this child's genome adds



The lead mechanism and the patient-specific checks (Section 7.1).

Evidence tiers: **A** = mechanism shown in a BubR1 model and paediatric human safety data; **B** = mechanism in a BubR1 model, limited paediatric data; **C** = indirect rationale only.

#	Agent	Mechanistic link	Key evidence	Status and paediatric safety	Main concerns	Tier
1a	**Nicotinamide riboside (NR)**	NAD^+ precursor → SIRT2 deacetylation of BUBR1 K668 → less degradation	NAD^+ precursor (NMN) raised BubR1 in mice [5]	Supplement. Open-label studies in ataxia-telangiectasia: n=24, mean age 17.5 [8]; 2-year study, n=10 incl. 7 children aged 3+ [9]; case starting at 3.5 years, no adverse effects [10]	Partial rescue at best; theoretical concern that raising NAD^+ could support tumour growth in a cancer-prone child	**A**
1b	**Niacin (nicotinic acid)** — *approved medication*	NAD^+ precursor (Preiss–Handler route) → same SIRT2 → BUBR1 axis	Raised blood NAD^+ up to 8-fold in humans and restored muscle NAD^+ in mitochondrial myopathy [26]; exploratory LINCS: 98th of 5,425 reversers (Section 4)	Prescription medicine for dyslipidaemia; long clinical use; paediatric experience limited	Flushing; liver toxicity at high or sustained-release doses; switches off 3 alarm genes (incl. *CDKN1A*) in LINCS, to be monitored	**B**
2	Nicotinamide mononucleotide (NMN)	Same axis	Directly tested: increased BubR1 in vivo [5]	Little paediatric data; regulatory status varies by country	As NR, with less human safety data	**B**
3	Nicotinamide (vitamin B3)	NAD^+ precursor	By analogy with NR/NMN	Approved, inexpensive, long clinical use	At high concentrations it *inhibits* sirtuins, which could work against the goal	**B/C**
4	N-acetylcysteine (NAC)	Antioxidant	Mice with BUBR1 at ~20% of normal had higher mortality under oxidative stress [11]; no antioxidant was tested	Approved, long paediatric use	Weak, indirect link; exploratory only	**C**
5	Senolytic (dasatinib + quercetin)	Removes senescent cells that accumulate after missegregation	*Genetic* clearance of $p16^+$ senescent cells delayed ageing disorders in BubR1 hypomorphs [12]; D+Q itself not tested there	Dasatinib is a cytotoxic kinase inhibitor	Toxicity likely outweighs benefit. **Reserve only**	**C**

Lead strategy: NAD^+ repletion, with two options. Niacin is the *approved medication* on this axis: a prescription vitamin B3 drug that raised blood NAD^+ up to 8-fold in a human trial [26]. **Nicotinamide riboside** has the stronger *paediatric* safety record (open-label use in children with ataxia-telangiectasia from age 3) and causes no flushing; as nicotinamide riboside chloride it received FDA orphan-drug and rare paediatric disease designations for ataxia-telangiectasia in June 2024 (manufacturer announcement), but it is sold as a supplement, not an approved medicine. Both act on the defect the mechanism identifies (BUBR1 turnover), via the NAD^+ → SIRT2 lever shown in a BubR1 mouse model [5]. The choice between them, and whether to use either, belongs to the clinical team after the cell experiments (Section 10).

What NAD^+ repletion cannot do. Even SIRT2 overexpression did not restore normal BubR1 levels in mice [5, 6]. The realistic aim is to **reduce the rate of missegregation**, not to cure the condition.

7.1 Personalised checks from the child's genome

Two local checks (`tools/check_vcf_genes.py`, `tools/check_pgx.py`) asked whether *this* child can use the proposed drugs and which future drugs carry extra genetic risk for him. Only gene-level categories are reported; genotypes stay local.

Can he convert the lead drugs into NAD^+? None of the nine genes needed showed a rare protein- or splice-altering variant: the niacin route (*NAPRT*, *NADSYN1*), the NR route (*NMRK1*, *NMRK2*), the shared final step (*NMNAT1–3*), the salvage enzyme (*NAMPT*) and the target lever itself (*SIRT2*). **His genome supports both options**, so the choice between niacin and NR rests on tolerability and clinical judgement.

Pharmacogenomic safety (19 sites in 10 genes).

Gene	Drugs affected	Result for this child
CEP72	Vincristine	**Risk genotype present**: associated with more severe vincristine-induced neuropathy in children [28]
CYP3A5	Vincristine	**Non-expresser**: slower vincristine clearance, associated with more neuropathy [29]
CYP2C19	PPIs, voriconazole, clopidogrel	Intermediate metaboliser: follow CPIC guidance if these are used
TPMT, *NUDT15*, *DPYD*, *UGT1A1*, *G6PD*, *CYP2C9*, *SLCO1B1*	Thiopurines, fluoropyrimidines, irinotecan, rasburicase, NSAIDs, statins	No tested risk alleles

Why this matters. Vincristine is standard chemotherapy for the tumour type seen in MVA, and the patient's muscle feature was classified as possibly treatment-related (Section 3.1). Carrying risk factors at two vincristine loci gives a biological reason for heightened vulnerability, and is a flag for the oncology team if vincristine is used again. These are research-grade associations without a formal dosing guideline; they inform monitoring, not a change in treatment by themselves.

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8. What this could and could not help

MVA type 1 typically involves slow growth before birth, small head size, developmental differences, and a raised risk of childhood cancer [15]. The proposed mechanism acts only on **cells that are still dividing**, so its potential reach is limited and should be stated plainly.

Feature (general MVA picture)	Driven by	Could stabilising BUBR1 help?
Ongoing chromosome errors in dividing tissues (e.g. blood)	Current checkpoint weakness	**Possibly.** This is the direct target and the main measurable outcome.
Childhood cancer risk	Accumulating aneuploid cells over time	**Possibly, in the long term**; unproven. Tumour surveillance must continue unchanged.
Growth after birth	Partly ongoing cell division	**Uncertain**; at most a modest effect.
Growth restriction before birth, small head size, structural birth differences	Development that has already happened	**No.** A treatment started now cannot reverse these.
Developmental and neurological features	Mostly set during brain development	**Unlikely**; mature neurons do not divide.

The patient-specific phenotype analysis is summarised in Section 3; individual clinical features are not reproduced in this public report.

Drug and management considerations for the secondary features. These are not repurposing candidates for MVA, but they follow from the analyses above:

- **Muscle (vincristine susceptibility):** if vincristine is used, closer neuropathy monitoring; avoid strong CYP3A4 inhibitors such as azole antifungals alongside it (they raise vincristine exposure, which matters more in a *CYP3A5* non-expresser), preferring an echinocandin for antifungal cover where clinically appropriate [32].
- **Possible *PEX5* disorder:** only if the peroxisomal blood tests confirm it, oral **cholic acid**, approved as adjunctive treatment for peroxisomal disorders including Zellweger spectrum with liver involvement, would become relevant; nutritional support is standard either way.
- **Renal:** review loop-diuretic exposure and urinary calcium; standard nephrology management.
- **Clotting (Factor V Leiden):** awareness during central-line use; avoid oestrogen-containing medicines later in life.

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9. Notes for clinicians

These are considerations to support a clinical team's own judgement, not a protocol. Dosing, if any, must be decided by the treating clinicians.

Before any trial

- Establish a baseline: percentage of aneuploid cells in blood (karyotype or interphase FISH, or single-cell sequencing where available), blood NAD^+ metabolites, routine bloods.
- Confirm cell-level response first (Section 10, steps 1–2), ideally in the child's own cells.

During any trial

- Repeat the aneuploidy measurement at pre-set intervals; this is the primary readout.
- Continue the child's existing cancer surveillance exactly as before. Because raising NAD^+ could in theory support tumour cell metabolism, any new mass or unexplained change should prompt stopping and review.
- Use a pre-specified stop rule (adverse effects, rising aneuploidy, or no change after a defined period).
- If niacin is chosen: monitor liver tests and expect flushing; avoid sustained-release forms at high doses.
- Second-diagnosis work-up (Section 3.2): plasma very-long-chain fatty acids and related peroxisomal markers, Sanger confirmation of the *PEX5* variant, serum alpha-1 antitrypsin level.
- Family (Section 3.3): parental Factor V Leiden and *BUB1B* carrier testing; a recurrent-pregnancy-loss work-up for the mother (including antiphospholipid antibodies); genetic counselling.
- Pharmacogenomics (Section 7.1): if vincristine is used, the child's *CEP72* and *CYP3A5* results suggest closer neuropathy monitoring; *CYP2C19* intermediate metaboliser status is relevant to PPIs and voriconazole.
- Prefer NR or niacin over high-dose nicotinamide: at high concentrations nicotinamide inhibits sirtuins, the opposite of the intended effect.

Drug classes to use with caution in MVA generally Agents that disrupt cell division or damage DNA (Section 6) may worsen chromosome errors in a child whose checkpoint is already weak. Where alternatives exist for other indications, clinicians may wish to weigh this.

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10. Validation plan (pre-registered success and failure criteria)

Each step proceeds only if the previous one meets its success criterion; the thresholds are fixed now, before any data exist.

Step	Model and treatment	Primary readouts	Success (proceed)	Failure (stop or revise)
1. Cells	Patient-derived fibroblasts or lymphoblastoid cells (or *BUB1B*-depleted cells re-expressing the missense allele); NR and nicotinic acid across a dose range, 72 h; vehicle control	Cellular NAD^+; BUBR1 protein; BUBR1 K668 acetylation; missegregation (micronuclei, lagging chromosomes by live imaging)	$NAD^+ \geq 1.5\times$ vehicle **and** BUBR1 ≥ 25% higher **and** missegregation ≥ 25% lower	NAD^+ rises but BUBR1 does not: abandon the NAD^+ strategy
2. Specificity	As step 1, with *SIRT2* knockdown	BUBR1, missegregation	Effect lost without SIRT2	Effect persists without SIRT2: mechanism is not as proposed; re-examine before step 3
3. Safety signal	As step 1	p53/interferon alarm genes (*CDKN1A*, interferon-stimulated genes)	Alarm intact	Alarm suppressed (flagged for niacin in LINCS): prefer NR or stop
4. Mice	BubR1 hypomorphic mice, NR in diet	BubR1 level, aneuploidy in splenocytes, tumour incidence, survival	Lower aneuploidy, no rise in tumours	Any tumour increase: stop
5. Supervised n-of-1	Only with the child's clinicians and ethics oversight	Blood NAD^+ (weeks), % aneuploid blood cells by interphase FISH (months), liver tests, tumour surveillance	NAD^+ rise and stable or falling aneuploid fraction	Pre-set stop rule: adverse effects, rising aneuploidy, or any new tumour signal

11. Impact, innovation and scalability

- **Impact.** Both lead options are cheap and oral: niacin is an approved medicine and NR has paediatric safety data, so it could be tested quickly if the cell work is positive. The exclusion table is useful on its own: it tells clinicians which commonly used drug classes to be cautious with in MVA.
- **Innovation.** The strategy targets protein *stability* (the actual defect in this genotype class), not the gene. Three independent routes (gene, phenotype, transcriptome) are combined, each feature of the patient is classified as genetic or not with a network test, and the transcriptomic route shows that reversal must exclude drugs that silence the p53/interferon alarm. The pipeline combines direction-aware filtering, a whole-drug safety pass and explicit labelling of literature-only evidence.
- **Scalability.** Both tools are disease-agnostic: the gene-first screen runs on any gene via a JSON config, and the phenotype-first analysis runs on any patient's HPO list; the chromosome-instability hazard filter is shared across MVA genes (*BUB1B*, *TRIP13*, *CEP57*). Every run writes a full exclusion log and provenance file, so results are auditable and reproducible.

Reproduce:

```
 cd track2/pipeline
python3 repurpose.py --config config/bub1b.json --out results/
# 19 Oct 2026 run: 7 retained (6 literature + 1 for review), 551 excluded; live databases drift

cd ../phenotype
python3 phenotype.py --hpo patient_hpo.txt --gene BUB1B --related CEP57,TRIP13 \
                     --config ../pipeline/config/bub1b.json --out results/
# public demo: --hpo example_public_mva1.txt

cd ../lincs
python3 signature_reversal.py --genes BUB1B,CEP57,TRIP13 --config ../pipeline/config/bub1b.json --out results/
# expected: axis test p = 0.085; 93 of top 100 reversers switch off the alarm

# local only (never commit the output): second-diagnosis check
python3 ../tools/check_vcf_genes.py --vcf /path/to/patient.vcf.gz
python3 ../tools/check_vcf_genes.py --vcf /path/to/patient.vcf.gz --genes
NAPRT,NADSYN1,NMRK1,NMRK2,NMNAT1,NMNAT2,NMNAT3,NAMPT,SIRT2
python3 ../tools/check_pgx.py --vcf /path/to/patient.vcf.gz
python3 ../tools/check_feature_genes.py --vcf /path/to/patient.vcf.gz --hpo <unexplained HPO IDs> --skip-genes
BUB1B,CEP57,TRIP13 --details
```

12. Methods and AI use

- **Data used:** the hackathon case data (Track 1 genotype; patient HPO terms, processed locally) and public resources (DGIdb, Open Targets, HPO, STRING, LINCS L1000 via Enrichr, WHO ATC via ChEMBL, published literature). No subject data are reproduced in this report or the repository.
- **Code:** `track2/pipeline/repurpose.py` (gene-first screen) and `track2/phenotype/phenotype.py` (phenotype-first analysis; patient HPO terms are read from a local, git-ignored file), `track2/lincs/signature_reversal.py` (transcriptome-first route) `track2/tools/check_vcf_genes.py` (local gene-panel checks: second diagnosis, NAD^+ conversion genes) , `track2/tools/check_pgx.py` (local pharmacogenomic and family-history check) and `track2/tools/check_feature_genes.py` (local

genome-wide second-diagnosis scan); configs in `track2/pipeline/config/`; every run writes a provenance file recording sources, configuration and gene set.
- **AI assistance:** Anthropic, Claude Opus 5.5 via claude.ai, Max plan; "Help improve Claude" setting **off**. Used for literature triage, drafting text and writing pipeline code; all references and outputs were checked by the team. The patient's HPO term IDs were entered into the assistant; no VCF, clinical document or other case files were uploaded, and the second-diagnosis VCF check was run locally.

13. Limitations

- The variant effects are predictions; protein levels in this child's cells have not been measured.
- Direct evidence that NAD^+ raises BUBR1 comes from mice; human evidence covers NAD^+ raising and safety, not BUBR1 itself (tested first, Section 10). Mouse and human BubR1 may differ in regulation; the human K668 equivalence should be confirmed.
- NR safety data come from a different disease and small, open-label studies.
- Database results depend on DGIdb and Open Targets at run time; counts can change as these update.
- LINCS signatures come from cancer cell lines and measure transcripts, not protein stability; NR and NMN are not in LINCS.
- Niacin was not in the pre-specified LINCS hypothesis; its strong reversal score is exploratory, and the post-hoc group p-value (0.024) is not corrected for that choice.
- The genome-wide scan covers coding exons and canonical splice regions of known disease genes only; deep-intronic, structural and copy-number variants were not assessed, and two heterozygous variants cannot be phased without parental data.
- The genetic-or-not classification uses HPO annotations, which are incomplete; "secondary" and "acquired" are hypotheses pending the clinical history.

Future work. Align the raw reads (not used here) to review copy number across the tubulopathy genes (especially *CLCNKB* deletions) and *CYP2D6*, if the clinical history does not explain the renal features; test niacin, NR and the exploratory division-restoring compounds side by side in patient cells (Section 10).

References

1. Suijkerbuijk SJE, et al. Molecular causes for BUBR1 dysfunction in the human cancer predisposition syndrome mosaic variegated aneuploidy. *Cancer Res.* 2010;70(12):4891–4900.
2. Hanks S, et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in *BUB1B*. *Nat Genet.* 2004;36(11):1159–1161.
3. Elowe S. Bub1 and BubR1: at the interface between chromosome attachment and the spindle checkpoint. *Mol Cell Biol.* 2011;31(15):3085–3093.
4. Suijkerbuijk SJE, et al. The vertebrate mitotic checkpoint protein BUBR1 is an unusual pseudokinase. *Dev Cell.* 2012;22(6):1321–1329.
5. North BJ, et al. SIRT2 induces the checkpoint kinase BubR1 to increase lifespan. *EMBO J.* 2014;33(13):1438–1453. PMID 24825348.
6. Cosentino C, Mostoslavsky R. Sirtuin to the rescue: SIRT2 extends life span of BubR1 mice. *EMBO J.* 2014;33(13):1417–1419.
7. Baker DJ, et al. BubR1 insufficiency causes early onset of aging-associated phenotypes and infertility in mice. *Nat Genet.* 2004;36(7):744–749.
8. Veenhuis SJG, et al. Nicotinamide riboside improves ataxia scores and immunoglobulin levels in ataxia telangiectasia. *Mov Disord.* 2021. doi:10.1002/mds.28788.
9. Presterud R, et al. Long-term nicotinamide riboside use improves coordination and eye movements in ataxia telangiectasia. *Mov Disord.* 2024;39(2):360–369. doi:10.1002/mds.29645. PMID 37899683.
10. Nicotinamide riboside for ataxia telangiectasia: a report of an early treated individual. *Neuropediatrics.* 2023;54(1):78–81. doi:10.1055/a-1959-9404.
11. Matsuda D, et al. BUBR1 insufficiency in mice increases their sensitivity to oxidative stress. *In Vivo.* 2016;30(6):769–776. PMID 27815460.
12. Baker DJ, et al. Clearance of p16Ink4a-positive senescent cells delays ageing-associated disorders. *Nature.* 2011;479(7372):232–236.
13. Freshour SL, et al. Integration of the Drug–Gene Interaction Database (DGIdb 4.0) with open crowdsource efforts. *Nucleic Acids Res.* 2021;49(D1):D1144–D1151.
14. Ochoa D, et al. The next-generation Open Targets Platform. *Nucleic Acids Res.* 2023;51(D1):D1353–D1359.
15. MedlinePlus Genetics. Mosaic variegated aneuploidy syndrome. U.S. National Library of Medicine. https://medlineplus.gov/genetics/condition/mosaic-variegated-aneuploidy-syndrome/
16. Gargano MA, et al. The Human Phenotype Ontology in 2024: phenotypes around the world. *Nucleic Acids Res.* 2024;52(D1):D1333–D1346.

17. Szklarczyk D, et al. The STRING database in 2023. *Nucleic Acids Res.* 2023;51(D1):D638–D646.
18. Subramanian A, et al. A next generation connectivity map: L1000 platform and the first 1,000,000 profiles. *Cell.* 2017;171(6):1437–1452.
19. Xie Z, et al. Gene set knowledge discovery with Enrichr. *Curr Protoc.* 2021;1(3):e90.
20. Bakhoum SF, et al. Chromosomal instability drives metastasis through a cytosolic DNA response. *Nature.* 2018;553(7689):467–472.
21. WHO Collaborating Centre for Drug Statistics Methodology. ATC classification, group L01 (antineoplastic agents), accessed via ChEMBL.
22. Posey JE, et al. Resolution of disease phenotypes resulting from multilocus genomic variation. *N Engl J Med.* 2017;376(1):21–31.
23. Cheng J, et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science.* 2023;381(6664):eadg7492.
24. Rentzsch P, et al. CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Res.* 2019;47(D1):D886–D894.
25. Jaganathan K, et al. Predicting splicing from primary sequence with deep learning. *Cell.* 2019;176(3):535–548.
26. Pirinen E, et al. Niacin cures systemic NAD^+ deficiency and improves muscle performance in adult-onset mitochondrial myopathy. *Cell Metab.* 2020;31(6):1078–1090.
27. Karczewski KJ, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature.* 2020;581(7809):434–443.
28. Diouf B, et al. Association of an inherited genetic variant with vincristine-related peripheral neuropathy in children with acute lymphoblastic leukemia. *JAMA.* 2015;313(8):815–823.
29. Egbelakin A, et al. Increased risk of vincristine neurotoxicity associated with low CYP3A5 expression genotype in children with acute lymphoblastic leukemia. *Pediatr Blood Cancer.* 2011;56(3):361–367.
30. Clinical Pharmacogenetics Implementation Consortium (CPIC) guidelines, cpicpgx.org (accessed October 2026).
31. Rey E, et al. Thrombophilic disorders and fetal loss: a meta-analysis. *Lancet.* 2003;361(9361):901–908.
32. Moriyama B, et al. Adverse interactions between antifungal azoles and vincristine: review and analysis of cases. *Mycoses.* 2012;55(4):290–297.
33. Kim HS, et al. SIRT2 maintains genome integrity and suppresses tumorigenesis through regulating APC/C activity. *Cancer Cell.* 2011;20(4):487–499.
34. Shoji M, et al. Nicotinamide riboside supplementation benefits in patients with Werner syndrome: a double-blind randomized crossover placebo-controlled trial. *Aging Cell.* 2025. doi:10.1111/acel.70093.

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